XF

PORTFOLIO · 2024–2026

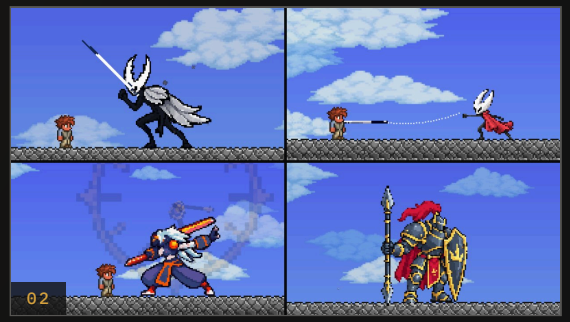
Xinyue Feng

ALSO KNOWN AS LILY

Games, mods & interactive web

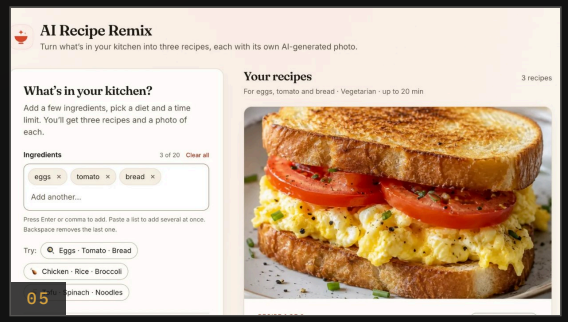
Computer Science student at Kean University, class of 2027. I make games and interactive web apps, and I care most about how they feel to use.

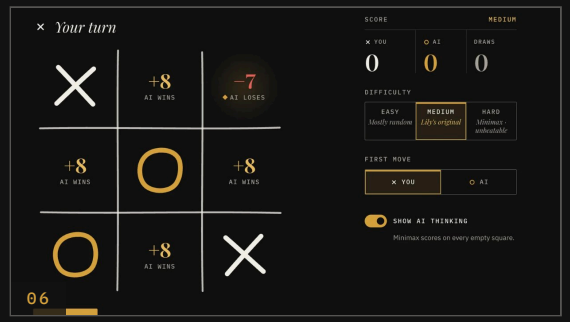












lily.china@outlook.com
samsi0322.github.io/xinyue-feng

About

I'm a Computer Science student at Kean University, graduating in 2027. I'm drawn to human–computer interaction and user-centered design, and I learn best by building things people can use and play.

The Java platformer I designed, drew and coded on my own grew out of a game I loved, *Ace Attorney*. I also built a recipe app that turns what's in your fridge into three recipes with photos, and a tic-tac-toe game with a rule-based AI opponent. In summer 2026 I built Wizarding World, a Harry Potter adventure for Terraria, with AI coding and image tools.

This fall I've been leading two bigger projects with AI tools: Yuzu Onsen Capybaras, a cozy pixel-art life sim you can play in the browser, and Otherworlds, a Terraria mod that brings in 38 bosses from five other games.

EMAIL
lily.china@outlook.com

WEBSITE
samsi0322.github.io/xinyue-feng

EDUCATION

Kean University — EXPECTED 2027
B.S. in Computer Science
Relevant coursework: HCI / UX, Software Engineering, Data Structures, Databases, Web Development.

EXPERIENCE

Chengdu Sheer Tianyi Game Technology — JUL 2026
TA Intern Engineer, Information Technology Department
Worked with Cascadeur 3D animation and its AI frame interpolation, and on a price-comparison crawler for third-party ticketing platforms.

Chengdu Sanying Technology — MAY – JUN 2026
Computer Vision Data Processing Intern
Organized image data and helped classify images as part of a computer-vision team.

Chengdu 2025 World Games Executive Committee — 2025
Sports Intern · Outstanding Intern
Coordinated competition logistics and communication between teams.

AWARDS & ACTIVITIES

Dean's Honor List, Kean University — SPRING 2026

The Honor Society of Phi Kappa Phi — 2025–2026
Founded in 1897, Phi Kappa Phi (ΦΚΦ) is the oldest and one of the most prestigious multidisciplinary academic honor societies in the United States. Membership is highly selective, recognizing exceptional academic excellence, with eligible seniors typically ranking in the top 10% of their class.

CiGA Game Jam — 2024
Team member in a 48-hour jam: gameplay design and interaction flow for a playable prototype, refined through rounds of playtesting.

Lanqiao Cup, Third Prize — 2024
National Software & IT Competition, Visual Arts Design track, Zhejiang division.

TOOLBOX

Languages
Java · JavaScript / TypeScript · HTML & CSS

Frameworks
React · Tailwind CSS · Node.js & Express · Java Swing

Tools
Git & GitHub · REST APIs, OpenAI API · Aseprite · AI coding agents (Claude Code, Codex) · AI image generation

BEYOND CODE

I've volunteered as a tutor for children in rural areas and helped organize inclusive sports activities for autistic young people.

Selected work

2024–2026



01 · PIXEL-ART LIFE SIM · 2026

Yuzu Onsen Capybaras

A cozy life sim in a hot-spring town. Play the winter demo in your browser.

04

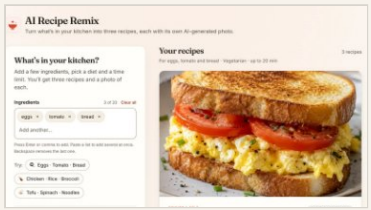


03 · TERRARIA MOD · 2026

Wizarding World

A Harry Potter adventure for Terraria: 12 boss fights and a Horcrux hunt.

11



05 · FULL-STACK WEB · 2026

AI Recipe Remix

Three recipes and a photo of each dish from what's in your kitchen.

16

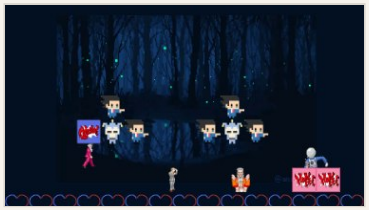


02 · TERRARIA MOD · IN DEVELOPMENT

Otherworlds

Thirty-eight bosses from five games come to Terraria with their signature moves.

08



04 · JAVA GAME · 2024

The Adventures of Edgeworth

An Ace Attorney platformer in Java: six levels, three kinds of enemy, one hidden block.

14



06 · GAME AI · 2026

Tic-Tac-Toe vs. AI

A rule-based AI and an unbeatable minimax search, side by side.

18

01

PIXEL-ART LIFE SIM · 2026

Yuzu Onsen Capybaras

A cozy life sim in a hot-spring town

Help out at a capybara pet shop in a little hot-spring town in the mountains: look after seven capybaras, grow a garden, make friends around town and join a festival every week, all year round.

ROLE
Lead developer, with AI tools

STATUS
Free winter demo in the browser; coming to Steam

PLATFORM
Browsers on computers and phones; Windows and macOS

BUILT WITH
JavaScript, HTML canvas, Web Audio, Electron; Claude Code and AI image generation

▶ Play the winter demo
capybara.sam-si.cn

Case study and trailer
samsi0322.github.io/xinyue-feng/work/yuzu-onsen-capybaras



A hot soak in the snow.



The shop and the town

The shop

Seven capybaras live at the shop, each with a personality of its own, and they get on with their day whether you help or not: napping, eating, paddling among the pool floats, soaking in the hot spring and sunbathing in the yard, with a little icon overhead to show how each one feels. Feed them, pat them, name them and dress them up while customers drop in, pay and leave their stars.

The town

Step out of the door and there's a whole town. Thirteen townsfolk keep days of their own, each with tastes, a birthday, letters and stories that unfold as you become friends, and six neighbors on Pine Lantern Street have stories too. There's a garden to grow, a forest to forage and fish in, a notice board of requests, an old bathhouse to bring back to life, and a book of wishes that changes the town as you fill it.

7 CAPYBARAS	**19** PEOPLE TO BEFRIEND	**16** FESTIVALS
18 CROPS	**30** CAVE FLOORS	**2** LANGUAGES



The shop at breakfast, with one capybara's menu open.

A year in town

Each of the four seasons has its own look, weather, crops, fish, recipes and sounds, and the townsfolk dress for each one. Every week brings a festival, sixteen in all:

WINTER
Snow Lantern Night, a capybara fashion show, a yuzu bath and New Year’s Eve.

SPRING
A doll festival on the stream, a blossom picnic, a spring fair with goldfish scooping, and a capybara race on Children’s Day.

SUMMER
Tanabata wishes, blindfolded watermelon splitting, fireworks over the pond and paper lanterns on the stream.

AUTUMN
Moon viewing, catching falling leaves, a harvest festival with a prize for the finest vegetable, and a sweet-potato bonfire.



The garden in summer, grown from one bed to three.



Spring: the blossom picnic.



Summer: fireworks over the pond.



Autumn: the leaf viewing in the park.



Winter: Snow Lantern Night in the square.

Under the mountain, and on your desktop



A floor of crystals in the Spring Vein.



The source, thirty floors down.



Goldfish scooping at the spring fair.



▶ Watch the 92-second trailer.

Under the mountain

Under the mountain shrine, the old spring keepers' tunnels go down thirty floors to the source of the town's hot water. You knock open rocks for ore, crystals and fossils, find each ladder by its wisp of steam, ride the steam lifts and fish the underground pools. There are no monsters: just the warm dark and your lantern.

The desk pet

A spin-off app puts one of the shop's capybaras on your desktop: it ambles along the bottom of the screen, and a toolbar gives it snacks, a hot bath, a paddling pool, a nap or the wardrobe. It runs on macOS and Windows.

Play it

Play the first winter free in your browser, on a computer or a phone: 28 days and all four winter festivals, with keyboard and mouse, touch or a controller, in English or Simplified Chinese. A desktop version with 20 achievements is coming to Steam.

Credits

Development led by Lily with AI tools: the code was written with Claude Code, the pixel art generated with AI image tools and the text written with the help of an AI language model. The music and sound effects are synthesized by the game as you play.

Pixel font: Fusion Pixel Font by TakWolf.

02

TERRARIA MOD · IN DEVELOPMENT

Otherworlds

A crossover boss mod for Terraria

Thirty-eight bosses from Hollow Knight, Hollow Knight: Silksong, Dead Cells, Cuphead and The Binding of Isaac come to Terraria with their signature moves. Each game arrives as its own pack of bosses, weapons and gear.

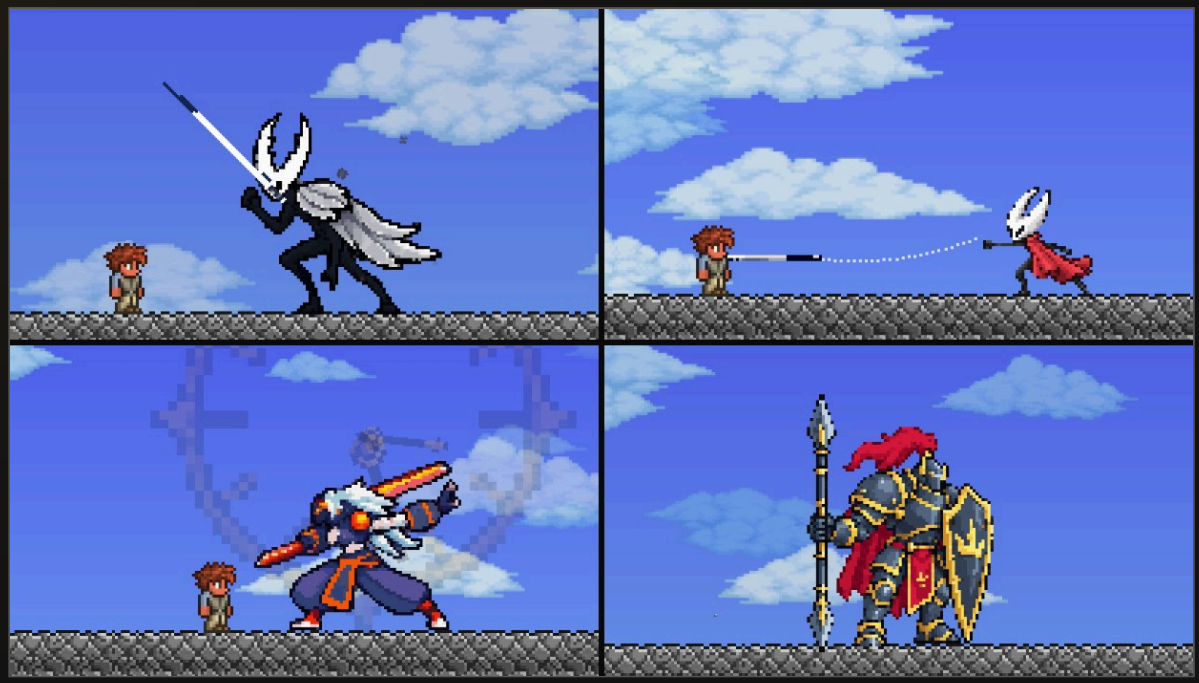
ROLE
Lead developer, with AI tools

STATUS
In development

PLATFORM
Terraria (tModLoader), single player and multiplayer

BUILT WITH
C#; Claude Code, OpenAI Codex and AI image generation

Case study
samsi0322.github.io/xinyue-feng/work/otherworlds



The Pure Vessel, Hornet, the Time Keeper and the Hand of the King.



Five games, thirty-eight bosses

The bosses fight with the moves they’re known for, and they’re spread across a whole playthrough, from the first days of a new world to beyond the Moon Lord. Their weapons and gear fit Terraria’s melee, ranged, magic and summoner classes.

38	80+	5	2
BOSS FIGHTS	WEAPONS	GAME PACKS	LANGUAGES

Hollow Knight · 13
From the False Knight to the Absolute Radiance

Silksong · 7
From the Bell Beast to the Bell Eater

Dead Cells · 7
From Mama Tick to the Queen

Cuphead · 9
From the Root Pack to the Devil

The Binding of Isaac · 2
Monstro and the Duke of Flies, down a trapdoor in a top-down basement

What you fight with

The nail, Soul that heals, spells and charms from Hollow Knight; Silksong’s crests and tools; Dead Cells-style weapons and skills; and Cuphead’s finger-gun shots, dash and parry of anything pink.

Eight of the thirty-eight



The Absolute Radiance (Hollow Knight).



The Pure Vessel focuses.



Grand Mother Silk (Silksong).



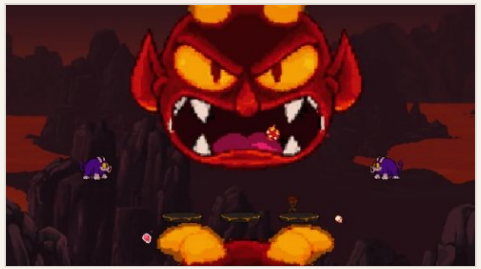
Conjunctivius (Dead Cells).



The Time Keeper lunges.



Sally Stageplay (Cuphead).



The Devil's giant head.



Monstro in the basement (Isaac).

Down the trapdoor

A trapdoor opens onto an otherworld of its own: a top-down basement in Isaac's style, laid out fresh every run, with lit rooms, darkness between them and no gravity. Monstro and the Duke of Flies wait below, and beating them earns keepsakes you can use back up in Terraria.

Summon, fight, collect

Summon any boss when you're ready, then collect its treasure bag, trophy, mask and relic. Play alone or with friends, in English or Simplified Chinese.

Credits

Development led by Lily with AI tools: the code is written with Claude Code and OpenAI's Codex, and the sprites are generated with AI image tools.

An unofficial fan project, to be released free.

03

TERRARIA MOD · 2026

Wizarding World

A Harry Potter mod for Terraria

Start as a Hogwarts student with your first wand, fight through 12 bosses, hunt down the Horcruxes and face Lord Voldemort, then claim the Deathly Hallows.

ROLE

Solo developer, with AI tools: design, code and art direction

STATUS

1.0 release candidate, July 2026

PLATFORM

Terraria (tModLoader)

BUILT WITH

C#; Claude Code and AI image generation

Source and player guide

`github.com/SamSi0322/WizardingWorld`

Case study

`samsi0322.github.io/xinyue-feng/work/wizarding-world`



Hogwarts at night, under the Deathly Hallows.



What it is

Wizarding World turns Terraria, the 2D sandbox game, into a Harry Potter adventure, from your first wand to the final battle. It all unfolds in a Pensieve: relive memories from across wizarding history, from the Basilisk and Aragog to the Battle of Hogwarts, and replay any boss you like.

Between the bosses there's a whole wizarding world: wand lore with 18 woods, 5 cores and 9 spell schools, a Dark Arts meter, class quests, Quidditch seasons with Seeker ranks, Hogsmeade and a three-zone Forbidden Forest.

12 BOSS FIGHTS	**24** WANDS	**78** ENEMIES
16 TOWN NPCS	**64** ACCESSORIES	**3** LANGUAGES



Platform Nine and Three-Quarters.

Twelve bosses


Mountain Troll


Basilisk


Fluffy


Umbridge


Bellatrix


Dementor King


Quirrell


Aragog


Horntail


Fenrir


Barty Crouch Jr.


Lord Voldemort

The world, and the final battle



Hogwarts across the Black Lake.



The Quidditch pitch under a stormy sky.



The Forbidden Forest after dark.

Every area has its own backdrop, from Hogwarts across the Black Lake to the Forbidden Forest at night.

Before the final battle, destroy four Horcruxes and hunt down Nagini to weaken Voldemort. After him come the Deathly Hallows: claim the true Invisibility Cloak from Dumbledore, purify Gaunt's Ring into the Resurrection Stone, and equip all three Hallows to become Master of Death. A Boss Compass and a Wizard's Almanac show you where to go next.

Developed by Lily with AI tools: the code was written with Claude Code and the sprites generated with AI image tools. Harry Potter © Warner Bros. Entertainment and J.K. Rowling; Terraria © Re-Logic. A non-commercial fan project.



The poster: a wand raised under the Deathly Hallows.

04

JAVA GAME · SPRING 2024

The Adventures of Edgeworth

An Ace Attorney platformer

A Mario-style platformer in Java starring Ace Attorney's Miles Edgeworth: six hand-designed levels, three kinds of enemy, a hidden block and an ending. Play it in your browser.

ROLE
Solo: game design, level design, sprites and code (2024). The 2026 browser port and Java update were built with AI coding tools.

TIMELINE
Apr 26 – May 12, 2024

CONTEXT
CPS 2231 course project, Wenzhou-Kean University

BUILT WITH
Java 8, Swing, Java Sound · Eclipse, Aseprite, Excel

▶ Play in the browser
samsi0322.github.io/xinyue-feng/play/edgeworth

Case study
samsi0322.github.io/xinyue-feng/work/edgeworth



Edgeworth takes on Winston Payne and Manfred von Karma in the dark forest.

Courtroom regulars, blocks and hearts

The idea

Lily built the game around two things she loved: the little monster-adventure games on the 4399 web-game site she played as a child, and Capcom's *Ace Attorney*. So the hero is the prosecutor Miles Edgeworth, the enemies are courtroom regulars, and the blocks are the series' "Hold it!" and "Objection!" speech bubbles.



Miles Edgeworth walks, jumps, stomps.



Winston Payne paces; land on his head.



Manfred von Karma can't be stomped.



Wendy Oldbag rises out of pipes.



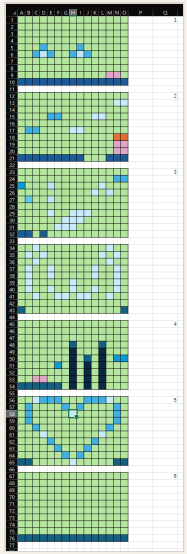
Dick Gumshoe points at a hidden block.

How it works

Nine classes, about 1,400 lines, no game engine: a double-buffered 900 × 600 window, and the player, every enemy and the end flag each on its own thread. Collision works on the tile grid:

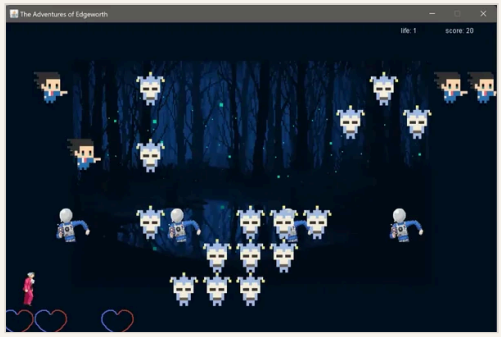
```
//Cannot move to the right
if(ob.getX()==this.x+60 &&
        (ob.getY()+50>this.y && ob.getY()-50<this.y)){
    if(ob.getType() != 3){
        canRight = false;
    }
}
```

From Edgeworth.java. Type 3 is the hidden block: you can walk through it until it's revealed.

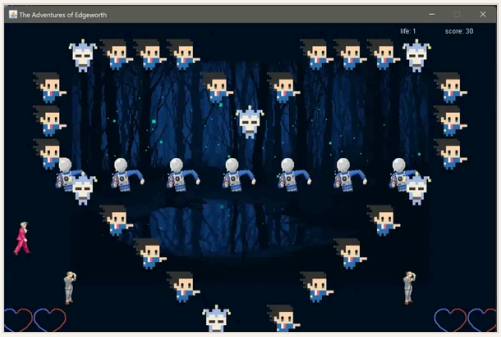


The level plans in Excel.

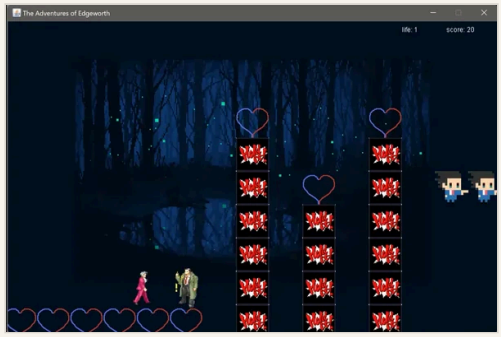
Every scene is a 15 × 10 grid of 60-pixel tiles, planned in a spreadsheet, one colored cell per block. Scene 3 is a face, scene 5 a heart, and scene 4's pillars are too tall to climb until Gumshoe's hidden "Objection!" block appears.



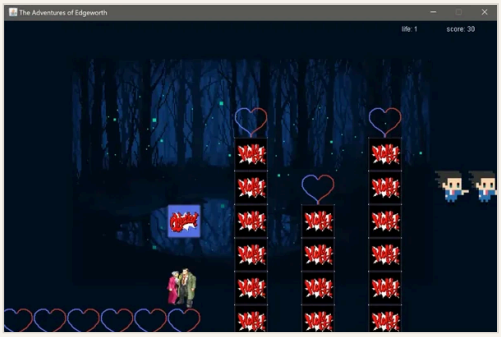
Scene 3, "the face."



Scene 5, a heart.



Before: Gumshoe waits.



After: the climb begins.

Game, levels, sprites and original code: Lily. 2026 browser port and Java update: built with AI coding tools. Characters from Capcom's *Ace Attorney*, in a non-commercial fan game; forest animation by Anas Abdin.

05

FULL-STACK WEB · FEBRUARY 2026

AI Recipe Remix

From leftovers to plated recipes

A web app that turns whatever is in your kitchen into three recipes, each with a photo of the finished dish.

ROLE

Solo: front end, back end and prompts. The 2026 portfolio edition (new design, demo mode and structured output) was built with AI coding tools.

TIMELINE

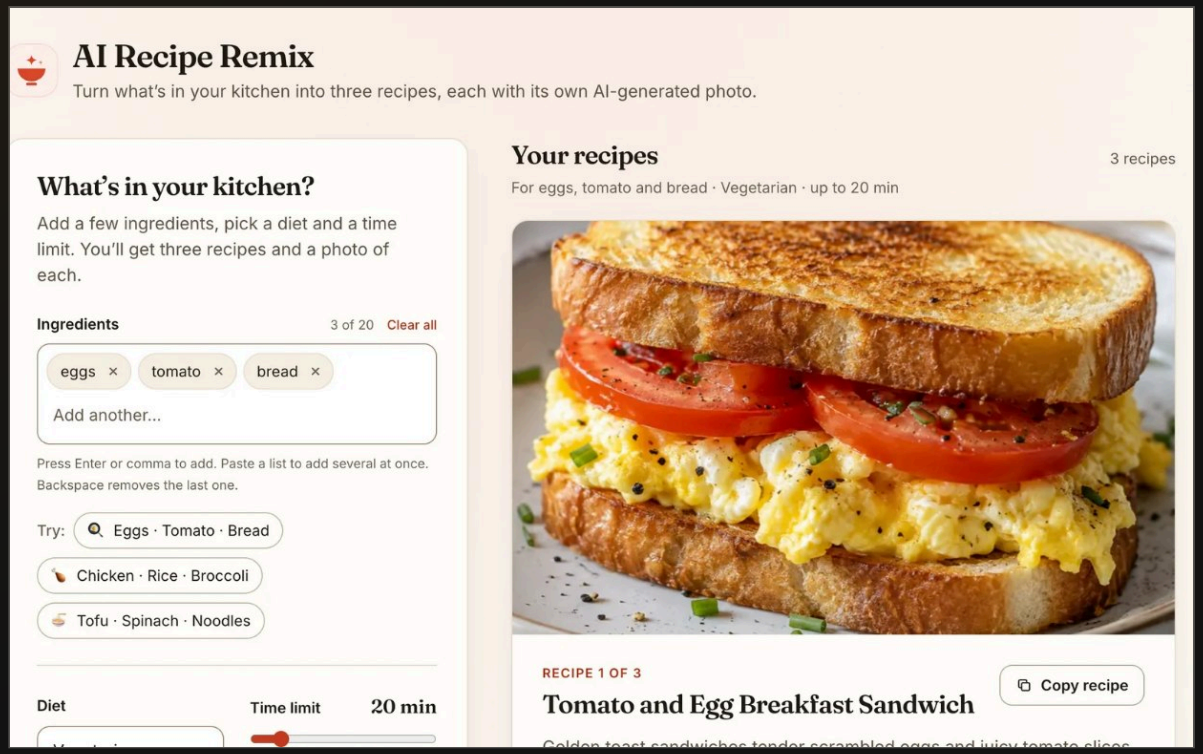
February 2026

BUILT WITH

React, Vite, Tailwind CSS, Express, OpenAI API

Try the demo
samsi0322.github.io/xinyue-feng/play/recipe-remix

Case study
samsi0322.github.io/xinyue-feng/work/recipe-remix



Eggs, tomato and bread become a breakfast sandwich, plus two more recipes.

From the fridge to the plate



Skillet Tomato Eggs with Toast Soldiers



Garlic Soy Chicken and Broccoli Rice Bowls



Tofu, Spinach, and Mushroom Noodle Soup

The idea

Most recipe sites start from a dish; this one starts from your fridge. Type what you have, choose a diet and a time limit, and it suggests three realistic recipes with quantities, steps, substitutions, allergens and tips, plus what you'd still need to buy.

How it works

Recipes and photos come from two separate requests, so the recipe cards appear within seconds and each photo fades in when it's ready. The Express server asks the model for JSON in a fixed shape and keeps the OpenAI key private.

Try it

The demo on the website plays back three sample kitchens. To cook up new recipes, run the app with your own OpenAI key.

Three dishes from the demo, each with the photo the app made for it.

06

GAME AI · APRIL 2026

Tic-Tac-Toe vs. AI

Three opponents, one unbeatable

A browser game with three computer opponents: one that plays loosely, one that follows five fixed rules, and one that searches ahead to the end of the game and never loses.

ROLE
Solo: the original game and its rule-based AI. The 2026 portfolio edition (Easy and Hard, “Show AI thinking” and a new design) was built with AI coding tools.

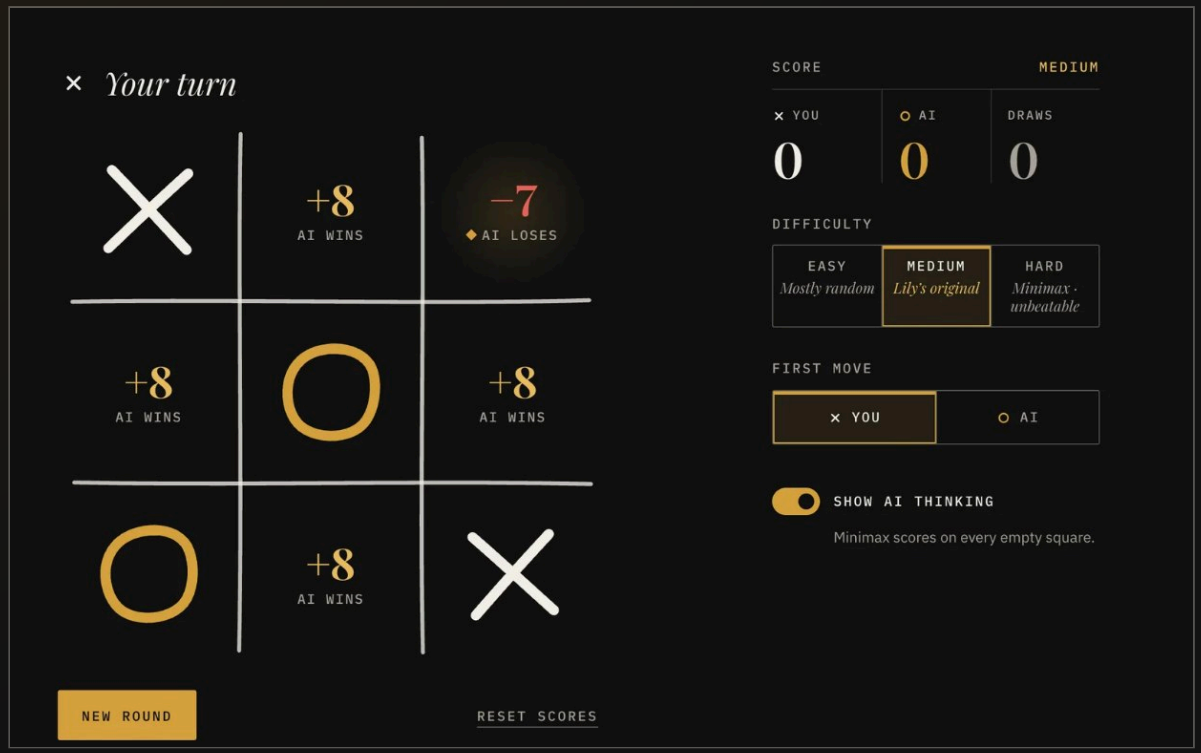
TIMELINE
April 2026

CONTEXT
Unit 10 class activity, “Play with AI”

BUILT WITH
HTML, CSS, JavaScript; no libraries

▶ Play
samsi0322.github.io/xinyue-feng/play/tic-tac-toe

Case study
samsi0322.github.io/xinyue-feng/work/tic-tac-toe



“Show AI thinking” scores every empty square.

Rules versus search

Her rule-based AI

Lily's opponent plays by five rules, checked in order:

1. Win if it can.
2. Block you if you're about to win.
3. Take the center.
4. Take a free corner.
5. Otherwise, take a side.

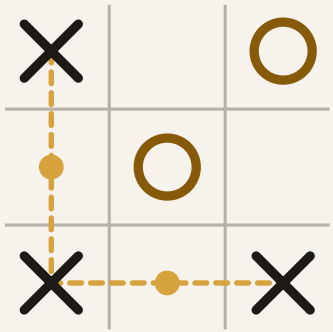
It plays sensibly and can be beaten: take a corner, then the opposite corner, and your block creates a fork it can't answer.

Three opponents

Easy mostly picks a random free square, but pounces on a win about half the time.

Medium plays by her five rules.

Hard uses minimax: it searches to the end of the game, assumes you always play your best, and picks the best guaranteed result. Alpha-beta pruning keeps each search to a few milliseconds.



The fork: two lines each need one more X, and Medium can only block one.

Seeing the AI think

With "Show AI thinking" on, every empty square shows the minimax verdict for it: a forced win, a draw or a square that would let you win. You can watch a move turn from draw to loss the moment you set up a fork.

16,752

Possible games against Hard. It loses none.

CONTACT

Say hello

lily.china@outlook.com

samsi0322.github.io/xinyue-feng

PLAY

Yuzu Onsen Capybaras, winter demo
capybara.sam-si.cn

The Adventures of Edgeworth
samsi0322.github.io/xinyue-feng/play/edgeworth

Tic-Tac-Toe vs. AI
samsi0322.github.io/xinyue-feng/play/tic-tac-toe

AI Recipe Remix demo
samsi0322.github.io/xinyue-feng/play/recipe-remix

Wizarding World on GitHub
github.com/SamSi0322/WizardingWorld

This portfolio, its website and the 2026 updates to the older projects were built with AI coding tools. *Ace Attorney* © Capcom; *Harry Potter* © Warner Bros. Entertainment / J.K. Rowling; *Terraria* © Re-Logic; *Hollow Knight* and *Silksong* © Team Cherry; *Dead Cells* © Motion Twin; *Cuphead* © StudioMDHR; *The Binding of Isaac* © Edmund McMillen.